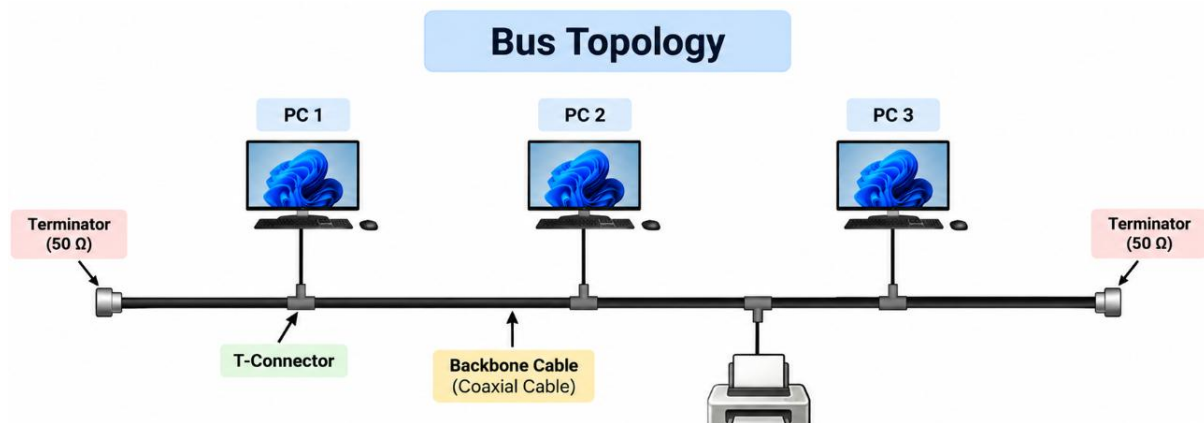


Network Topologies Assignment

Task – 1 : Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

Solution : Bus Topology



Key components

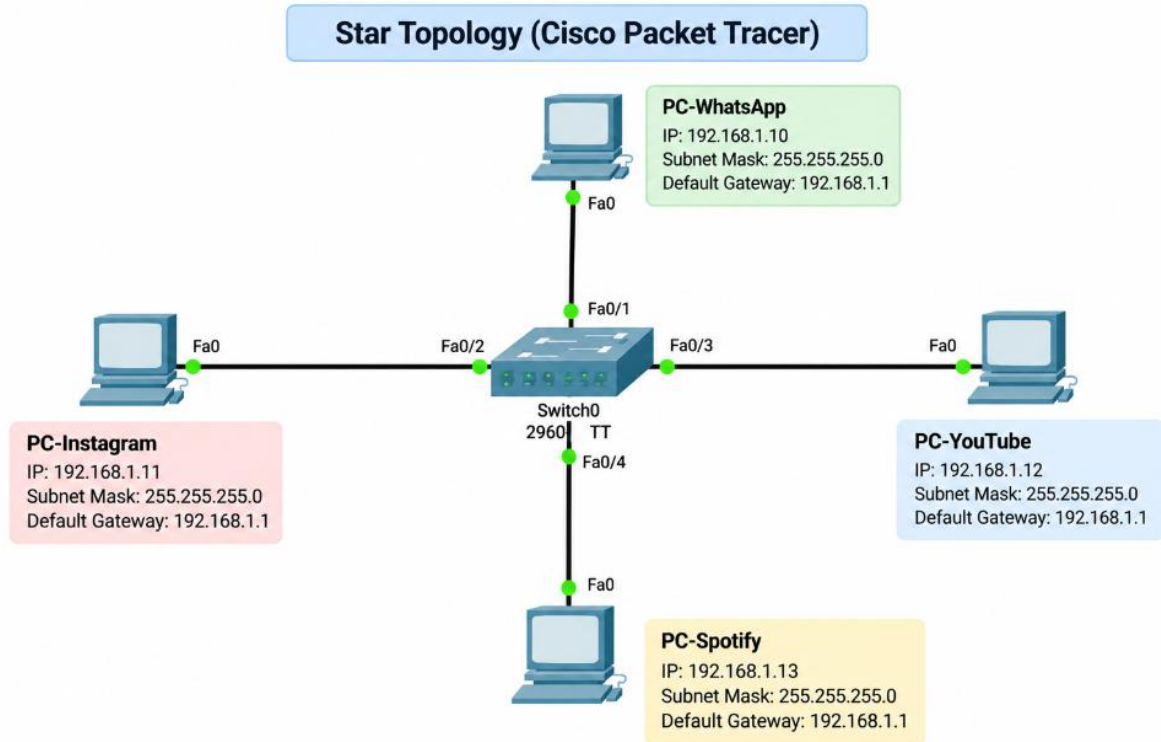
- **Backbone cable:** Main cable through which data travels.
- **Nodes/Devices:** PCs or other network devices connected to the backbone.
- **Terminators:** Placed at both ends of the cable to prevent signal reflection.
- **Connectors:** Used to connect devices to the backbone.

Real-world example

A bus topology was traditionally used in **older Ethernet networks using coaxial cables**, such as 10BASE2 and 10BASE5 networks. Modern networks generally use star topology instead because it is easier to manage and troubleshoot.

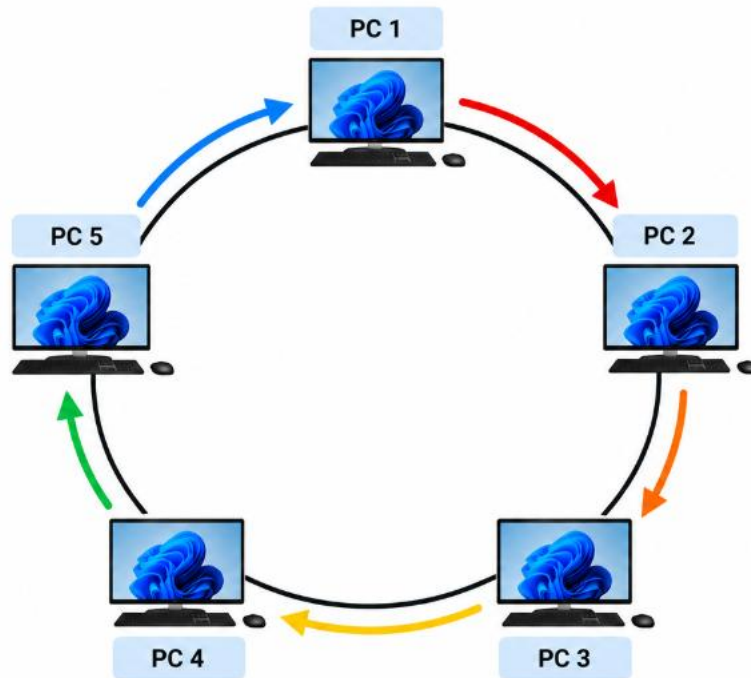
Task – 2 : In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

Solution :



Task – 3 : Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup. Hint : Use arrows to show the direction of data flow.

Solution : Ring Topology



How data travels

In a ring topology, each device is connected to two neighboring devices, forming a closed loop. Data travels from one device to the next around the ring until it reaches the intended destination. Depending on the implementation, data may travel in **one direction or both directions**.

Task – 4 : Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Solution : **Mesh vs Star Topology**

Feature	Mesh Topology	Star Topology
Advantage 1	Provides high reliability because devices can have multiple paths between them.	Easy to install, manage, and troubleshoot.
Advantage 2	Failure of one connection usually does not affect the rest of the network.	Failure of one cable/device generally affects only that individual device.
Disadvantage 1	Requires a large amount of cabling and many connections.	If the central switch/hub fails, the connected devices lose network connectivity.
Disadvantage 2	More expensive and complex to install and maintain.	Requires a central networking device and additional cabling.